Pneumatics

National 5 · Mechanisms & structures · Worksheet

ENGINEERING WITH MR KELLY

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ name pneumatic components and draw their symbols
- ☐ describe circuits with the correct terms: actuate, change state, outstroke, instroke
- ☐ identify AND and OR control in pneumatic circuits
- ☐ calculate force, pressure and area in N/mm²
- ☐ describe speed control and time delays with restrictors and reservoirs

I DO — Worked example

Read how a full-mark answer is built

A single-acting cylinder is controlled by a push-button 3/2 valve.

1 Describe the operation of the circuit. (3 marks)

Full-mark answer each ✓ is one mark

When the push button actuates the 3/2 valve, main air flows from port 1 to port 2 and the piston outstrokes ✓. When the button is released, the spring returns the valve ✓. Air in the cylinder exhausts through port 3 and the spring instrokes the piston ✓.

Why answers lose marks

"Press the button, the valve opens, air goes in and pushes the piston out. Let go and it goes back."

1 mark at most: "opens", "pushes out" and "goes back" aren't the accepted terms.

WE DO — Guided practice

Use the starters and word bank to help you

Two 3/2 valves are piped in series to control a cylinder.

2 State the type of control and describe why it is used. (2 marks)

Sentence starters

1. …
2. …

Word bank

actuated · AND control · valves · safety · two-handed · operator's

Tip: 2 marks = 2 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) The correct way to describe a 5/2 valve operating is...
☐ It actuates ☐ It changes state ☐ It outstrokes ☐ It opens

b) Two valves piped in series give...
☐ OR control ☐ AND control ☐ NOT control ☐ A time delay

c) To slow a piston smoothly you should restrict...
☐ The supply air ☐ The exhaust air ☐ The pilot air ☐ The reservoir

d) A 12 mm diameter piston: area is about...
☐ 113 mm² ☐ 452 mm² ☐ 37.7 mm² ☐ 144 mm²

4 Key terms: write the letter of the matching description next to each term.

1. **Single-acting cylinder**	____	**A** A restrictor feeding a reservoir before a pilot valve.
2. **Double-acting cylinder**	____	**B** 3 ports, 2 states; "actuated".
3. **3/2 valve**	____	**C** 5 ports, 2 states; "changes state"; controls a double-acting cylinder.
4. **5/2 valve**	____	**D** Restricts air flow in one direction only.
5. **Uni-directional restrictor**	____	**E** Air drives the piston both ways.
6. **Pneumatic time delay**	____	**F** Air outstrokes; a spring instrokes.

A cylinder operating at 0.5 N/mm² has a piston area of 800 mm².

5 Calculate the outstroke force. (2 marks)

A press cylinder must outstroke slowly and smoothly.

6 Describe how the circuit can be modified. (2 marks)

A double-acting cylinder is controlled by a 5/2 pilot-operated valve and two 3/2 valves.

7 Describe what happens when valve A is actuated. (2 marks)

Check and reflect

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?